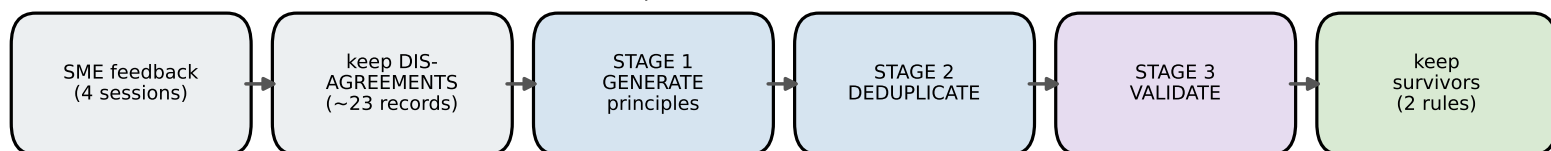


How ICAI Extracts — and Validates — Rules from SME Feedback

Inverse Constitutional AI: use the experts' accept/reject decisions to reconstruct the rules that explain them, then keep only rules that predict those decisions.



THE VALIDATION TEST — "reconstruction"

Idea: a good rule should let you REPLAY the experts' decisions — so we test whether the rule can PREDICT them.

1. Build a test set with KNOWN answers: the ~23 rejections (truth = "reject") + 20 accepts (truth = "accept"). Each case's truth = what the expert actually did.
2. Predict BLIND: give an LLM ONLY the rule + the requirement + the AI's proposed change (never the real decision) and ask, per case: reject / accept / na (not relevant).

3. Score STRATIFIED (the key step):

- corrective accuracy = of the rejections it covers, how many it correctly predicted "reject"
- accept accuracy = of the accepts it covers, how many it correctly predicted "accept"
- coverage = fraction of cases the rule is relevant to (verdict $\neq$ "na")

4. KEEP the rule only if: coverage $\geq$ 0.10 AND corrective $\geq$ 0.60 AND accept $\geq$ 0.50.

Why split them? A lazy rule that says "reject everything" scores corrective = 1.00 (looks flawless) but accept = 0.00 (it also rejects the accepts) — so the accept gate catches and drops it.

DROPPED ✗

"Prefer requirements that keep quantitative values"

corrective 1.00 (perfect!)
BUT accept 0.00 → over-broad

KEPT ✓

"Do not flag ... when specific, measurable values are present"

corrective 0.80, silent on accepts

The 2 rules that survived:

[A9] (backed by 13 expert decisions, corrective 0.8, coverage 0.349)
"Do not flag requirements for lacking traceability to stakeholder needs when specific, measurable values are already provided and necessary for clarity."

[A2] (backed by 5 expert decisions, corrective 1.0, coverage 0.349)
"Do not flag timing constraints when they are logically necessary for system performance, even if not explicitly traceable to stakeholder needs."

Caveat: within-sample test (tiny corpus, no held-out split) — it checks a rule coherently explains its own decisions without over-generalizing to accepts; a fresh reviewer (SME 4) would test true out-of-sample generalization.